

Applied Time Series Analysis.

Before learning any algorithms, let's understand what this course is about.

Many real-world datasets are collected over time. Examples include:

- Daily stock prices
- Monthly rainfall
- Hourly electricity consumption
- Weekly sales
- Daily COVID-19 cases

Unlike normal datasets, time is an important variable here. We cannot randomly shuffle the data because yesterday influences today.

This course teaches us how to analyze such data and make predictions about the future.

Time Series = Data collected over time.

Examples:

Data	Time
Temperature	Every day
Petrol Price	Every week
Gold Price	Every hour
Sales	Every month

Examples of data that changes every day?

- Temperature
- Petrol price
- Stock market
- Mobile data usage

Forecasting

Predicting future values using past observations.

Suppose you own a bakery.

Every day you prepare bread.

How many loaves should you bake tomorrow?

Too many?

They get wasted.

Too few?

Customers leave.

So, we need forecasting.

Real-life Examples

Hospital

Predict next week's patient count.

Amazon

Predict next month's sales.

IPL

Predict match attendance.

Restaurant

Predict tomorrow's customers.

If you were running a restaurant, what would you forecast?

- Customers
- Food demand
- Revenue

Forecasting $\neq$ Guessing

Forecasting uses Past data, Mathematics, Statistics

Why Forecasting?

"Imagine Amazon doesn't know tomorrow's demand.

What happens?

No stock.

Late deliveries.

Customer complaints.

Forecasting solves these problems."

Applications

Retail

Inventory planning

Hospitals

Medicine stock

Banks

Cash requirement

Weather Department

Rain prediction

Railways

Passenger demand

Airlines

Ticket pricing

Electricity

Power generation

Why can't companies simply wait until tomorrow?

Because planning requires advance knowledge.

Types of Forecasting

There are two broad types

Qualitative

Used when no historical data exists.

Example

Launching a new iPhone.

No previous sales.

Experts estimate demand.

Quantitative

Historical data available.

Use mathematics.

Examples

Stock prices

Sales

Electricity demand

Easy Example

Suppose your friend opens a café.

No past data.

Use expert opinion.

↓

Qualitative

After 3 years

Daily sales available.

↓

Quantitative

Qualitative Forecasting Methods

These depend on

Experience

Knowledge

Judgement

Examples

- Market Survey
- Expert Opinion

Real-life Example

Apple launching a new product.

No historical sales.

Experts estimate demand.

Can machine learning predict sales if there is no historical data?

No. Machine learning needs historical data.

Quantitative Forecasting

Historical data exists.

We use Statistics, Machine Learning, Time Series Models

Examples

- ARIMA
- Prophet
- Exponential Smoothing

Simple Analogy

Teacher predicting marks.

If previous tests exist

↓

Quantitative

No previous tests

↓

Qualitative

Time Series Forecasting

Suppose we know sales for the last 36 months.

Can we estimate month 37?

Yes.

That is Time Series Forecasting.

Predictor Variables

Some forecasts depend on other variables.

Example

Ice cream sales depend on

Temperature.

Electricity demand depends on

Weather.

Hospital admissions depend on

Pollution.

Time Series Models

Introduce Different forecasting models exist.

Examples

- Linear Regression
- Exponential Smoothing
- ARIMA
- Prophet
- VAR

We'll study each model in upcoming units.

Examples of Time Series

Daily Gold Price

Monthly Inflation

Yearly Population

Electricity Demand

Rainfall

Is your CGPA a Time Series?

No. Because it is not measured continuously over equally spaced time intervals.

Time Plot

Introduce the concept of a time plot, the most basic visualization in time series analysis.

Whenever we receive time series data, the first thing we should do is plot it.

A time plot helps us see how the data changes over time and often reveals important patterns before we perform any mathematical analysis.

For example, if we plot daily temperatures over one year, we may notice

- Higher temperatures during summer.
- Lower temperatures during winter.
- An overall upward trend if climate change is affecting the region.

A time plot is simply a graph where

- X-axis = Time
- Y-axis = Variable being measured

Real-Life Example

Imagine your bank balance recorded every day. Plotting it over time lets you see

- Whether your savings are increasing.
- Whether you spend more at the end of each month.
- Whether there are unusual spikes (e.g., salary credit).

Why Time Plots?

Graphs tell stories much faster than tables.

Suppose I show you this table

Month	Sales
Jan	120
Feb	130
Mar	145
Apr	160

Can you immediately tell whether sales are increasing?

Not as easily.

Now imagine a graph connecting those points. The increasing trend becomes immediately obvious.

Time plots help us:

- Understand data quickly.
- Detect trends.
- Detect seasonality.
- Find unusual observations.
- Identify missing data.

Which is easier to understand a table or a graph?